

well-known binary compression coding of the same type for the memoryless source p . Let us denote by $\rho_g^{(n)}$ and $\rho_c^{(n)}$ the generation ratio and the compression ratio for block length n , respectively. Then we can show immediately by the entropy branching property that $H(p) \leq \rho_c^{(n)} < H(p) + (1/n)(\log k(n) + 2)$, and $H(p) - (1/n)(\log k(n) + 2) < \rho_g^{(n)} \leq H(p)$, where

$$k(n) = \binom{n+r-1}{r-1}, \quad \log k(n) = O(\log n).$$

We see that the generation $p \rightarrow (1/2, 1/2)$ and the compression coding have the opposite objects: the former is intended to make the average number of output symbols per input symbol as large as possible under the condition that the output must obey $(1/2, 1/2)$ without unique decodability, whereas the latter is intended to produce as few symbols as possible with unique decodability but without regard to the output distribution. The entropy $H(p)$, however, is the ideal limit for both.

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The Second Order Reed-Muller Code of Length 64 Has Covering Radius 18

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Abstract—The assertion of the title is proved by first showing that the covering radius of $R(2, 6)$ cannot exceed 18 and then constructing a weight 18 coset leader.

Let $R(r, m)$ denote the r th order Reed-Muller code of length 2^m . We use the notation $|v|$ to denote the Hamming weight of a binary vector v . A coset leader of $R(r, m)$ is a binary vector v of length 2^m such that $|v+c| \geq |v|$ for all c in $R(r, m)$. The covering radius of $R(r, m)$, denoted $\text{CR}(r, m)$, is the weight of a coset leader of maximum weight among all coset leaders of $R(r, m)$. The purpose of this note is to prove that $\text{CR}(2, 6) = 18$. This computation completes the determination of $\text{CR}(r, m)$ for all r and m such that $0 \leq r \leq 6$ and $1 \leq m \leq 6$.

Lemma: If $1 \leq r \leq m$ then

$$\begin{aligned} 2\text{CR}(r, m-1) &\leq \text{CR}(r, m) \\ &\leq \text{CR}(r, m-1) + \text{CR}(r-1, m-1). \end{aligned} \quad (1)$$

Proof: Using the inductive definition of Reed-Muller codes described in [2, p. 374] we have

$$R(r, m) = \{(u, u+v) : u \in R(r, m-1), v \in R(r-1, m-1)\}.$$

Let (a, b) be an arbitrary binary vector of length 2^m where a and b are binary vectors of length 2^{m-1} . There exists a codeword $u \in R(r, m-1)$ such that $|a+u| \leq \text{CR}(r, m-1)$. Moreover, there is a codeword $v \in R(r-1, m-1)$ such that $|(b+u)+v| \leq \text{CR}(r-1, m-1)$. Hence there exists a vector in the coset

$(a, b) + R(r, m)$ whose weight is at most $\text{CR}(r, m-1) + \text{CR}(r-1, m-1)$. The right side inequality in (1) now follows. Finally, since $R(r-1, m-1)$ is a subcode of $R(r, m-1)$, if a is a coset leader of $R(r, m-1)$ then (a, a) is a coset leader of $R(r, m)$. The left side inequality in (1) follows and the proof of the lemma is complete.

Theorem: $\text{CR}(2, 6) = 18$.

Proof: We first show that $\text{CR}(2, 6) \leq 18$. In [3] McLoughlin proved that $\text{CR}(2, 5) = 6$ while the fact that $\text{CR}(1, 5) = 12$ follows from the Berlekamp-Welch analysis of the cosets of $R(1, 5)$ [1]. Thus, the lemma shows that $\text{CR}(2, 6) \leq 18$.

In order to prove that $\text{CR}(2, 6) \geq 18$, we will exhibit a specific coset of $R(2, 6)$ in which the minimum weight is 18. Using the notation of [2], we associate binary 2^m -tuples with Boolean functions of m variables by indexing the 2^m coordinate positions by the binary m -tuples in lexicographic order and regarding the entry in each position as the value of the Boolean function on the corresponding m -tuple. The code $R(2, 6)$ consists of the binary 64-tuples associated with the polynomials $f(v_1, \dots, v_6)$ of degree 2 or less in the variables $v_1, \dots, v_6$.

We now consider the binary 64-tuple associated with the polynomial

$$\begin{aligned} f(v_1, \dots, v_6) &= v_6(v_1v_2v_3 + v_1v_4v_5 + v_2v_3 + v_2v_4 + v_3v_5) \\ &\quad + (v_6 + 1)(v_1v_2v_3 + v_1v_4v_5). \end{aligned} \quad (2)$$

A simple computer program was written to generate the weight distribution of the coset $f + R(2, 6)$. If D_i denotes the number of vectors of weight i in the coset we find that

$$\begin{aligned} D_{18} &= D_{46} = 3584, \\ D_{22} &= D_{42} = 72192, \\ D_{26} &= D_{38} = 548352, \\ D_{30} &= D_{34} = 1473024, \end{aligned}$$

and $D_i = 0$ for all other i . Since this coset has minimum weight 18 it follows that $\text{CR}(2, 6) \geq 18$ and so we obtain $\text{CR}(2, 6) = 18$. This completes the proof.

The polynomial (2) is a natural candidate for a weight 18 coset leader of $R(2, 6)$ because $v_1v_2v_3 + v_1v_4v_5$ yields a weight 6 coset leader of $R(2, 5)$ and $v_1v_2v_3 + v_1v_4v_5 + v_2v_3 + v_2v_4 + v_3v_5$ yields a weight 12 coset leader of $R(1, 5)$. On the other hand, there are cases when equality holds in the left side inequality of the lemma. For example, Mykkeltveit [4] has proved that $\text{CR}(1, 7) = 56$, and the fact that $\text{CR}(1, 6) = 28$ is proved in [2, p. 426]. Hence, $2\text{CR}(1, 6) = \text{CR}(1, 7)$.

The Berlekamp-Welch study of the cosets of $R(1, 5)$ was useful in constructing the polynomial (2). Moreover, using their results it is possible to show that $f + R(2, 6)$ has minimum weight 18 without using the computer, but the proof is tedious.

The smallest Reed-Muller codes for which the covering radius is not known are $R(2, 7)$ and $R(3, 7)$. Since $\text{CR}(1, 6) = 28$, $\text{CR}(2, 6) = 18$, and by McLoughlin's work, $\text{CR}(3, 6) = 8$, the lemma above yields the bounds: $36 \leq \text{CR}(2, 7) \leq 46$ and $16 \leq \text{CR}(3, 7) \leq 26$.

We close by summarizing the known exact results on $\text{CR}(r, m)$. Since $R(m-1, m)$ consists of all even weight 2^m -tuples we have $\text{CR}(m-1, m) = 1$. And $R(m-2, m)$ is an extended Hamming code so $\text{CR}(m-2, m) = 2$. McLoughlin [3] has proved that

$$\text{CR}(m-3, m) = \begin{cases} m+2, & m \text{ even} \\ m+1, & m \text{ odd.} \end{cases}$$

When m is even it is known that $\text{CR}(1, m) = 2^{m-1} - 2^{(m-2)/2}$.

This result is due to Rothaus [5]. The only other exact values of $CR(r, m)$ known to this author are mentioned above, namely, $CR(1, 5) = 12$, $CR(1, 7) = 56$, and $CR(2, 6) = 18$.

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